

• GASTRIC CANCER •

Effects of Chinese Jianpi herbs on cell apoptosis and related gene expression in human gastric cancer grafted onto nude mice

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Supported by Shanghai High-Education Bureau Research Fund, No.
98QN72

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Received 2002-05-02 **Accepted** 2002-06-09

Abstract

AIM: To explore the mechanism of the Sijunzi decoction and another Chinese herbal recipe (SRRS) based mainly on the Sijunzi decoction in treatment of gastric cancer.

METHODS: A human gastric adenocarcinoma cell line SGC-7901 grafted onto nude mouse was used as the animal model. The mice were divided into 3 groups, one control and the two representative experimental conditions. Animals in the two experimental groups received either Sijunzi decoction or SRRS over a 40-day period starting at 1st day after grafting. Control animals received saline on an identical schedule. Animals were killed 41 days after being grafted. The effect of therapy was assessed by two ways: (1) tumor size was periodically measured during the life of the animals; (2) tumor weight was determined by a electron balance immediately after the animals killed. For detection of apoptotic cells, apoptotic indices(AI) were examined by the terminal deoxynucleotidyl transferase-mediated deoxyuridine triphosphate fluorescence nick end labeling (TUNEL) method. Morphological alterations were observed with electron microscopy. S-P immunohistochemical method was used to detect the expression of Ki-67 in xenografts. Expression of bcl-2 and p53 was semiquantitatively detected using a reverse transcriptase-polymerase chain reaction (RT-PCR) technique.

RESULTS: When compared with controls, tumor growth (size and weight) was significantly inhibited by treatment with the Sijunzi decoction ($P<0.05$) or SRRS ($P<0.01$). The tumor inhibitory rate in the Sijunzi decoction group was 34.33 % and SRRS group 46.53 %. AI of human gastric cancer xenografts in nude mice was significantly increased to 16.24 ± 3.21 % using TUNEL method and 11.38 ± 6.46 % by FACScan in the Sijunzi decoction group compared with the controls (TUNEL: 2.63 ± 1.03 %, $P<0.01$; FACScan: 7.15 ± 1.32 %, $P<0.05$). SRRS group was also found a significantly increased AI by using TUNEL method and flow cytometry analysis compared with the controls (TUNEL: 13.18 ± 3.05 %, $P<0.05$; FACScan: 11.58 ± 5.71 % ($P<0.05$). Under electron microscope, cell shrinkage, nuclear chromatin condensation,

formation of membrane blebs and apoptotic bodies were frequently observed in Sijunzi decoction group and SRRS group. The average labeling index (LI) for Ki-67 in SRRS group was significantly decreased to 8.43 ± 2.22 % compared with the control group (10.37 ± 4.91 %) ($P<0.05$). The average labeling index for Ki-67 in sijunzi decoction group was 7.95 ± 2.54 % which was lower than that of the control group, but showed no significance ($P=0.07$). The expression level of p53 mRNA was lower in both Sijunzi decoction group and SRRS group than that in control group ($P<0.05$; $P<0.01$). The expression of bcl-2 mRNA was also decreased in SRRS group compared with the control ($P<0.01$).

CONCLUSION: The inhibition of gastric cancer cell growth *in vivo* by Chinese Jianpi herbs and SRRS is related to induction of the cell apoptosis which may be involved in aberrant expression of p53 and bcl-2 genes

Zhao AG, Zhao HL, Jin XJ, Yang JK, Tang LD. Effects of Chinese Jianpi herbs on cell apoptosis and related gene expression in human gastric cancer grafted onto nude mice. *World J Gastroenterol* 2002; 8(5):792-796

INTRODUCTION

Apoptosis plays a crucial role in the proliferation and turnover of cells in various tumors. It has been clear that its extent is often enhanced in tumor by many anticancer drugs^[1-5], such as cytotoxic drugs^[6], hormone^[11], or some Chinese herbal medicine^[7-10]. In clinic studies, some Chinese Jianpi herbs had been proved to have effect on malignant tumors, especially on gastric and colorectal tumors^[11-13]. Among these herbs we found that Codonopsis pilosula (Franch) Nannf., Atractylodes macrocephala koidz. and the Sijunzi Decoction might suppress gastric carcinoma cell proliferation and cause tumor cell loss and the nuclear condensation *in vitro*^[14]. The Sijunzi Decoction and another Chinese herbal recipe SRRS of deheat-toxin, softening hard lumps and dissolving phlegm enhance apoptosis of human gastric cancer xenografts in nude mice^[15]. Based on previous studies, we examined the apoptotic indices of human gastric cancer grafted onto nude mice after the treatment with Sijunzi Decoction and SRRS and investigate the underlying mechanism of the tumor suppressive effect of these Chinese Jianpi herbs.

MATERIALS AND METHODS

Materials

Animal models Thirty 6-7 weeks old female BALB/C-nu/nu mice (weight 18-22 g) and a human gastric carcinoma cell line SGC-7901 were obtained from Shanghai Tumor Institute (No.01842). The animals were subcutaneously grafted with the SGC-7901 cell. The tumor transplantation procedure was described previously^[15].

Drugs The Sijunzi Decoction is composed of Codonopsis pilosula (Franch) Nannf., Atractylodes macrocephala koidz, Poria cocos (Schw.) Wolf, Glycyrrhiza uralensis Fisch. The concentration of the Sijunzi Decoction was 160 g/L; SRRS is composed of Atractylodes macrocephala koidz., Poria cocos(Schw.) Wolf, Sargentodoxa cuneata(Oliv.) Rehd. Et Wils., Prunella vulgaris L. And etc. The concentration of the SRRS decoction was 240 g/L.

Experimental schedule After grafting the mice were randomly divided into 3 groups, one control and the two experimental groups assigned to receive the Sijunzi Decoction or SRRS. Each animal in the two experimental groups was given 0.5 mL of the Sijunzi Decoction or SRRS by gastric perfusion every day over a 40-day period beginning at 1st day after grafting. The control animals received normal saline according to the same schedule. Animals were killed 41 days after being grafted.

Methods

Tumor growth The effect of therapy was assessed by two ways: (1)tumor size was measured twice a week by multiplying two perpendicular diameters. (2)tumor weight was determined immediately by electron balance after the animals were killed.

Apoptosis For detection of apoptotic cells, apoptotic indices were examined by the terminal deoxynucleotidyl transferase-mediated deoxyuridine triphosphate fluorescence nick end labeling (TUNEL) method^[16-18] and flow cytometry analysis. Morphological alterations were observed with electron microscope. (1)TUNEL: *In situ cell death detection Kit POD* (ISCCD, BOEHRINGER MANNHEIM) was used to detect the apoptotic cell. The procedures was according to protocol of the kit and the other references. The positive cells were identified, counted and analyzed under the light microscope. Non-necrotic zone was selected in the tissue section and images were sent to computer by AEC camera (Grundig Electronic Co. Ltd., Germany). 10 image at least 1000 cells were selected on the screen, positive ratio analyzed by KS400 Video Image Digital Analysis System (ZEISS, Germany). (2)Electron microscopy: Some of specimens in each group were fixed with 2.5 % glutaraldehyde. Semi-thin and ultra-thin sections were cut and viewed with scanning electron microscope. The characteristics of cell apoptosis showed nuclear chromatin condensation, peripheral masses of condensed chromatin with enclosed membrane or crescent. The nuclear membrane is complete. There is little or no swelling of mitochondria or other organelles. (3) Flow cytometry analysis: Propidium iodide (PI) staining^[19-21] was used for flow cytometric detection of

apoptosis. 10⁶ cells from each of the sample were treated with RNase and stained with PI. The apoptotic cells having DNA strand breaks that had been labeled were measured on a flow cytometer (FACSCalibur, Becton Dickinson, USA). The data from 10⁶ cells/sample were collected, stored, and analyzed using CELLQUEST (Becton Dickinson USA) and ModFIT LT for mac V1.01 software (Becton Dickinson, USA).

Cell proliferation The level of expression of Ki-67 was used as a marker of cell proliferation. In present study Ki-67 was measured by S-P immunohistochemical method.

Expression of bcl-2 and p53 mRNA The Expression of p53 (5-8 exons) and bcl-2 was semiquantitatively detected using RT-PCR technique^[22-24]. A β -actin was used as an internal standard. Sequences of the primers used for RT-PCR analysis are described in Table 1. Total cellular RNA was isolated from tumor tissues using the acid-guanidium-phenol-chloroform technique. The RNA extracted from each sample was qualified by agarose gel electrophoresis and ethidium bromide staining and the amount of RNA was determined by electrophotometry A:260/280. The extracted RNA was converted to first strand cDNA with AMV reverse transcriptase (Promega). p53 and β -actin were amplified by polymerase chain reaction consisting of stage one: 2 minutes of denaturation at 94 °C before addition of Taq DNA polymerase; stage two: 1 minute of denaturation at 94 °C, 1 minute of primer annealing at 56 °C, and 1 minute of extension at 72 °C (Taq DNA polymerase, Shanghai Sangon Biology Engineering Technique Service Co. Ltd.). Gene expression of bcl-2 was also analyzed by RT-PCR, in the same manner except for annealing at 55 °C. The cyclor of PCR gene amplification is from Omn-E (Hybaid). Each reaction tube contained: 1.5 μ l 25mM MgCl₂ +2.5 μ l 10 $\times$ PCR buffer +0.5 μ l 10mM dNTP +1 μ l $\times$ 2 20 pmol/ μ l p53 or bcl-2 primers or β -actin primers (primer concentration: 0.8 μ M) +2 μ l cDNA +0.5 μ l Taq (2.5-5U/ μ l) +16 μ l ddH₂O. Quantitative analysis: After the amplification step was completed, equal amounts (5 μ l) of PCR produces were loaded onto each lane of 1.7 % agarose gels and electrophoresed. FR-200UV/WHITE ANALYSIS (Shanghai Fu Ri Science & Technology Co. Ltd., China) took the image of electrophoresis with β -actin as internal standard, and FR-980 biological electrophoresis analysis system (Shanghai Fu Ri Science & Technology Co. Ltd., China) was used to analyze the quantity of nuclear acid.

Statistical analysis

The results were expressed as $\bar{x} \pm s$ and significant difference was assessed by Student's *t* test.

Table 1 Sequences of primers for amplified cDNA of the p53,bcl-2 and β -actin

Primers		Sequences	Combining sites	Amplifiers
p53	Sense	5'-GGAGGTTGTGAGGCGCTGC-3	645bp-663bp	311bp
	Antisense	5'-CACGCACCTCAAAGCTGTTC-3	936bp-955bp	
bcl-2	Sense	5'-CAGCTGCACCTGACGCCCTT-3'	1810bp-1829bp	191bp
	Antisense	5'-GCCTCCGTATCCTGGATCC-3'	2021bp-2040bp	
β -actin	Sense	5'-AGCGGGAAATCGTGCCTGAC-3	665bp-674bp	471bp
	Antisense	5'-ACTCCTGCTTGCTGATCCACATC-3	1103bp-1125bp	

Table 2 Chinese Jianpi herbs-induced effects on gastric cancer cell SGC-7901 ($\bar{x} \pm s$)

Treatment	Tumor weight / g (n)	Tumor size / mm ³ (n)	Percentage of control / %	Apoptotic index(AI) / %		ki-67 / %
				TUNEL (n)	FACScan (n)	
Sijunzi Decoction	0.66 $\pm$ 0.16(9) ^a	371.81 $\pm$ 52.51(9) ^a	34.33	16.24 $\pm$ 3.21(9) ^b	11.38 $\pm$ 6.46(9) ^a	7.95 $\pm$ 2.54(9) ^c
SRRS	0.54 $\pm$ 0.23(9) ^b	322.66 $\pm$ 126.14(9) ^b	46.53	13.18 $\pm$ 3.05(8) ^a	11.58 $\pm$ 5.71(8) ^a	8.43 $\pm$ 2.22(8) ^b
Control	1.01 $\pm$ 0.32(10)	603.61 $\pm$ 263.39(10)		2.63 $\pm$ 1.03(10)	7.51 $\pm$ 1.32(10)	10.37 $\pm$ 4.91(10)

^aP<0.05, ^bP<0.01, ^cP=0.070 vs control *t* test.

RESULTS

Chinese Jianpi herbs-induced effects on tumor growth

Each one case that the xenograft emerged later was eliminated in Sijunzi decoction group and SRRS group. Compared with the control group, tumor growth (size and weight) was significantly inhibited by treatment with the Sijunzi decoction ($P<0.05$) or SRRS ($P<0.01$). The tumor inhibitory rate of the sijunzi decoction group was 34.33 % and that in the SRRS group was 46.53 % (Table 2).

Chinese Jianpi herbs-induced effects on tumor cell apoptosis

Apoptotic index (AI) of xenografts in nude mice was significantly increased to 16.24 ± 3.21 % using TUNEL method and 11.38 ± 6.46 % FACScan in the Sijunzi decoction treatment group, compared with the controls (TUNEL: 2.63 ± 1.03 %, $P<0.01$; FACScan: 7.15 ± 1.32 %, $P<0.05$). SRRS group was also found a significantly increased AI by using TUNEL method and flow cytometry analysis compared with the controls (TUNEL: 13.18 ± 3.05 %, $P<0.05$; FACScan: 11.58 ± 5.71 %, $P<0.05$). But there was no significant difference between Sijunzi decoction group and SRRS group by using either TUNEL method or flow cytometry analysis. Under electron microscope cell shrinkage, nuclear chromatin condensation, formation of membrane blebs and apoptotic bodies were frequently observed in Sijunzi decoction group and SRRS group (Table 2).

Chinese Jianpi herbs-induced effects on tumor cell proliferation

The average labeling index for Ki-67 (LI) in SRRS treatment group (8.43 ± 2.22 %) was significantly lower than that in the control group (10.37 ± 4.91 %) ($P<0.01$). The average labeling index for Ki-67 in sijunzi decoction group was 7.95 ± 2.54 % which was lower than that of the control group, but showed no significance as the P value was 0.07 (Table 2).

Chinese Jianpi herbs-induced genetic effects

Expression of p53 (exons 5-8) and bcl-2 was semiquantitatively detected with β -actin used as an internal standard. The expression level of p53 mRNAs was 0.36 ± 0.27 , significantly lower in Sijunzi decoction group than that in control group (0.69 ± 0.20) ($P<0.05$), in SRRS group (0.19 ± 0.18) also significantly lower than that in control group ($P<0.01$). The expression of bcl-2 mRNA in SRRS group (0.33 ± 0.23) was decreased, compared with the control (0.81 ± 0.40) ($P<0.01$). In Sijunzi decoction group the expression of bcl-2 (0.41 ± 0.15) slightly decreased, compared with the control, but no statistics difference existed ($P=0.071$) (Table 3, Figure 1).

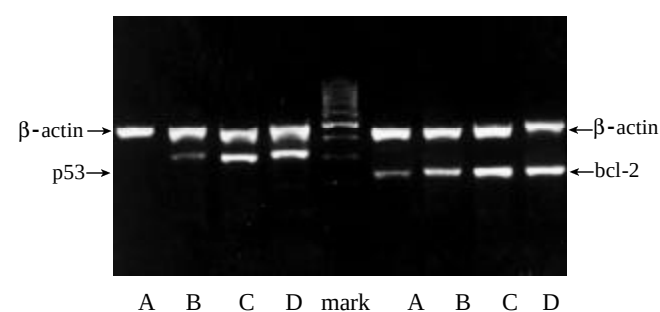


Figure 1 p53,bcl-2 mRNA expression in SGC-7901 grafted onto nude mice after treatment of Chinese Jianpi herbs by RT-PCR analysis. Lane A: SRRS group, Lane B: Sijunzi decoction group, Lane C: control group, Lane D: control group.

Table 3 Chinese Jianpi herbs-induced genetic effects ($\bar{x}\pm s$)

Treatment	p53(exons 5-8)	bcl-2
Sijunzi Decoction	$0.36\pm0.27(8)^a$	$0.41\pm0.15(8)^c$
SRRS	$0.19\pm0.18(9)^b$	$0.33\pm0.23(9)^b$
Control	$0.69\pm0.20(9)$	$0.81\pm0.40(9)$

^a $P<0.05$, ^b $P<0.01$, ^c $P=0.071$ vs control t test.

DISCUSSION

Despite its declining incidence, the gastric carcinoma remains one of the most common cause of cancer-related death in the world^[25-27]. At present gastric carcinoma is still detected later in most patients throughout the world, and even with curative resection, they remain at a high risk of relapse. Thus, there is a great need for effective adjuvant therapy for patients with gastric carcinoma^[28-30]. Our previous clinic paired comparative studies suggested that Chinese herbal recipe SRRS have therapeutic effects on advanced gastric cancer, with increasing the surviving period of the patients, improving the life quality, and decreasing the metastasis and recurrence rates after operation^[12,14]. Because of its lower toxic side-effect compared with chemical therapy, it is worth to make a further research on its anti-cancer mechanism.

As the other malignant tumor, gastric carcinoma is not only a disease with abnormal cell proliferation and differentiation, but also a disease with abnormal apoptosis^[5, 31-35]. The enhanced induction of apoptosis in human gastric carcinoma cells can be observed after treatment with 5-Fluorouracil^[36], Cisplatin^[37], arsenous oxide^[38], etc. These data suggest that inducing cancer cell apoptosis may be a therapeutic method for gastric carcinoma. The present study indicated that tumor growth was significantly inhibited by treatment with the Sijunzi decoction or SRRS. TUNEL method and cytometry analysis clarified that Sijunzi decoction and SRRS enhanced apoptosis. The results suggest that the inhibition of gastric cancer cells *in vivo* by Jianpi herbs described here is related to inducing apoptosis. Immunohistochemical staining for Ki-67 showed that SRRS inhibited cell proliferation. So the inhibition of gastric cancer by SRRS is also related to suppressing the proliferation.

Apoptosis is a complex, tightly regulated, and active cellular process whereby individual cells are triggered to undergo self-destruction in a manner that will neither injure neighboring cells nor elicit any inflammatory reaction^[1, 39-42]. Various triggering factor initiate corresponding proteolysis cascade reaction depending on mitochondrion or APO-1/FAS/CD95 receptor mediate apoptotic pathways^[40,43]. There are many oncogenes and tumor suppressor gene products in the regulation and execution of apoptosis. Among them are p53, Rb, myc, ras, raf, etc^[23, 40, 41, 44, 45]. p53, because of its role in apoptosis, has earned the name "guardian of the genome". It monitors the state of DNA, and in case of DNA damage, stalls the cell cycle. This takes place through the induction of CIP/WAF1/p21. In the absence of phosphorylated, active cyclin-dependent kinases, also another regulator of the cell cycle, Rb, remains inactive(unphosphorylated), and, hence, the cell cycle halts. This then leads to activation of DNA repair machinery. If the DNA repair fails, p53 takes over again and triggers apoptosis in a process that involves upregulation of the apoptotic-inducing bax and down-regulation of the apoptotic bcl-2^[23]. p53 also upregulates KILLER/DR5, a 45-kd apoptosis-inducing member of the tumor necrosis factor receptor family. Analogous to the APO-1/FAS/CD95 receptor

system, its activation also lead to a caspase activation^[37]. Thus p53 is known as one of the essential genes for cells to undergo apoptosis. In gastric cancer, mutant p53 expression decreased cancer cell apoptosis, p53 mutant provided selective growth superiority to the tumor cell. In our investigation we adopt the human gastric cancer cell line SGC-7901 which has point mutation in exon 6 codon 204 GAG→GCG coding Glu→Ala, which expresses mutant p53^[46,47]. RT-PCR detected p53 mRNA level was significantly lower in Sijunzi decoction group than that in control group. The expression level of p53 mRNA in SRRS group was also significantly lower than that in control group. The data suggest that these two recipes composed of Jianpi herbs induce SGC-7901 cancer cell apoptosis by down-regulation of mutant p53 mRNA expression.

We also detected apoptosis-inhibiting member of the bcl-2 family^[41,44]: bcl-2 mRNA. The expression of bcl-2 mRNA was decreased in SRRS group, compared with the control. In Sijunzi decoction group the expression of bcl-2 was slightly decreased compared with the control, but there was no statistics difference. It suggested that SRRS could suppress bcl-2 expression, but it was not stronger for than that of Sijunzi decoction. bcl-2 is the epitome of an antiapoptotic or survival gene. Attesting to its role in an apoptosis checkpoint, it counteracts apoptosis initiated by quite disparate signals, such as chemotherapeutic drugs, oxidative stress, viral infections, and p53. In gastric carcinoma, bcl-2 over-expressed at both protein and mRNA level in many cases^[48].

In our study SRRS and Sijunzi decoction down-regulated the expression of p53 mRNA and SRRS also decrease the expression of bcl-2 mRNA in SGC-7901 cancer cell undergoing apoptosis. We inferred that p53 and bcl-2 may be involved in the regulation of these Chinese Jianpi herbs inducing gastric cancer cell SGC-7901 apoptosis. The interaction contact of p53 and bcl-2 in SRRS or Sijunzi decoction inducing apoptosis needs further investigation.

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Edited by Zhang JZ